

smartphone applications are attempting to use this technology to produce high-frequency sounds to repel mosquitoes and other insects.

Ionic repellers. The concept of radio waves or radio frequency (RF) to control the behaviour of living organisms has shown promise. Flow of RF energy through free water molecules in the body of pests and insects generates heat due to the agitation of water molecules through ionic conduction of heat.

As a result, more thermal energy is produced inside pests and insects. Therefore RF treatment for pests and insects control, without negatively affecting food items, may serve as a desired alternative to chemical (that is, using pesticides) treatments, not only for household pests and insects but also for post-harvest pest control of different commodities. This can reduce long-term impact on environment, human health and competitiveness of agricultural industries.

Electromagnetic repellers. Electromagnetic pest control uses wiring within walls, turning the whole home or office into a giant pest repeller by sending a shifting signal through the wires, tuned to irritate the pests' nervous system. This technology does not affect humans, pets or electronics, or increase the wiring's own electromagnetic energy. It simply changes the electromagnetic frequency to target small pests, which easily pick up these changes in electromagnetic frequencies.

Signal-generating electromagnetic frequencies are regularly turned on and off every few seconds, so that pests cannot build tolerance towards electromagnetics. As insects communicate through vibrations, electromagnetic pest control vibrations communicate with pests, disturbing their receptors and making them feel like they are in a hostile area, driving them out as a result. Electromagnetic pest control, hence, makes it difficult or impossible for pests to gather food, breed, build nests or communicate with each other.

Electric current. Systems used in the electrical treatment of weed control are of two main types: spark discharges and continuous contact. The former uses high-voltage, short-duration pulses (for example, 25kV - 60kV, 1µs - 3µs), whereas the latter uses an

electrode connected to a high-voltage source (for example, 15kV, 54kW). As it touches plants, current flows for the duration of the contact time.

Plant tissue is damaged by current flow and shockwave of discharge in pulsed systems, and by rapid heating effect of electric currents in continuous contact devices. Commercial continuous contact equipment is available all over the world, for farm weed control, forest undergrowth clearance and railway track weed clearance.

Experimental systems have been developed for weed beet control (15kV rms, 50kW). These can treat infestation densities of up to 5000 stems per 10,000 square metres (one-hectare). Electrical weed control is versatile and rapid, and is cost-competitive with chemical applications if used on larger areas.

Pesticides and weedicides can be harmful to humans, if these come in contact with the body

Recently, use of weak electric current, particularly from photovoltaic devices, has become popular in keeping unwanted animals away from their agriculture fields. This technique has its own limitations in terms of infrastructure and cost. Also, it is ineffective in killing or repelling pests or controlling selective weeds already existing inside fields.

State of technology

Presently, electronic pest control products use an advanced, state-of-the-art microprocessor (just like a computer) to send ultrasonic waves, electromagnetic pulses and valuable negative ions to areas where pests rest, nest and breed, effectively forcing them out and making them run away. This technology is not only effective at driving pests out but is absolutely safe for people and biodiversity.

Electronic pest control devices are user-friendly, and are available for homes and offices in different models and at affordable costs. Some also come with air-purification facilities. However, the need of the hour is to develop these

electronic pest control technologies for their further effectiveness, cost-effectiveness and usefulness for large areas. These must have equal effect on pests and selective weed control in lawns, grasses and agriculture fields.

Bottlenecks of technology. Studies have described ultrasound pest control devices as ineffective, uneconomical and potentially harmful to users. Claims of effectiveness of electronic pest control, electric weed-control applications and ultrasonic control of pests are under debate.

Following are the points that must be taken into account while making use of electronic pest/weed control technology:

Keep trapping. Best results come from using ultrasonic pest repellents along with trapping.

Diminishing returns. Pests and weeds become accustomed to control techniques over time, so results may only be temporary.

Realistic expectations. Research on different electronic pest control devices has shown a decline, in addition to ineffective elimination of rodents, insects and pests.

Device location. Effective areas of electronic pest control devices are short and weak, and these are easily blocked by furniture and walls. It is important to extend their capacity.

Upgraded technology. Electronic pest control devices need to be equipped with powerful electronic chips and sensors, which can repel pests and rodents harder than before and work well on all types of pests.

Large coverage area. Almost all electronic pest control devices are effective only for an area up to 74.3sqm ~ 111.4sqm (800sqft ~ 1200sqft). This must be taken care of in the future.

Selective weed control. Electric weed control devices must be suitable for selective weed control/elimination, safe to humans, animals and crops/grasses, and powerful, cost-effective and safe.

Other requirements. All electronic pest control and electric weed control devices should be easy to use and cause no harm to humans, pets, crops, fruits, food and so on. These devices should act only when these are asked to. **EFY**